
VAC PRESENTATION

science of natural disasters and
their management

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Natural Disasters

Science and
management strategies
for resilience



Natural hazards analysis integrates physical science with operational planning to reduce loss.

This presentation outlines disaster *types*, assessment methods, and management actions to build **community resilience**.

Introduction

Types of Disasters



Meteorological hazards

Storms, hurricanes, heatwaves and severe convective events result from atmospheric dynamics.

Forecasting, attribution science, and early warning systems enable timely evacuations and preparedness.

Earthquakes, volcanoes, landslides and tsunamis originate from tectonic or slope instability processes.

Monitoring, seismic networks, and land-use planning are critical to reduce **structural risk**.

Geological hazards



Hydrological hazards

Floods, flash floods and droughts arise from hydrologic extremes and catchment responses.

River modeling, drainage management, and reservoir operations mitigate impacts and support water security.

Hazard Assessment



Spatial identification of hazard likelihood and exposure informs zoning and resource allocation.

Integrating hazard maps with population and infrastructure data supports **prioritized interventions**.

Risk mapping

Vulnerability analysis

Assessing social, economic, and infrastructural susceptibilities reveals where impacts will be greatest.

Targeted strengthening, social programs, and adaptive planning reduce long-term vulnerability.

Early warning systems

Early warning integrates hazard monitoring, data analysis, and communications to enable timely action.

Robust systems combine **real-time sensors**, predictive models, and multi-channel alerts to reduce response time and save lives.

Preparedness Strategies



Local planning aligns land use, evacuation routes, and critical infrastructure resilience with risk maps.

Engaging stakeholders in *participatory planning* ensures plans are practical and socially accepted.

Community
planning

Emergency training

Regular drills, role definition, and cross-agency exercises build operational readiness and reduce confusion during events.

Training should include vulnerable groups and scenario-based evaluations to measure competency.

Resource prepositioning

Strategic stockpiling and pre-deployment of supplies, equipment, and personnel shorten response timelines.

Logistics planning and supply chain resilience are essential for sustained operations.



Response & Recovery

Unified command structures and interoperable communication platforms enable efficient multi-agency responses.

Clear incident objectives and continuous situational awareness drive effective decision making.

Incident
coordination

Damage assessment

Rapid, standardized assessments quantify impacts on infrastructure, services, and populations to prioritize actions.

Use remote sensing and field teams to accelerate data collection and inform resource allocation.

Rehabilitation planning

Recovery plans restore services while integrating risk reduction and resilience upgrades.

Long-term financing, policy alignment, and community engagement support sustainable rehabilitation.

Integrated science, planning, and operations reduce disaster impact and accelerate recovery.

Prioritize **early warning**, targeted preparedness, and coordinated response to build resilient communities.

Conclusions

Q-1. Global warming and climate change are two major concerns in modern times. Explain how science and technology are trying to address these two issues.

Ans - Global warming and climate change represent a shift in the Earth's energy balance. To address this, science and technology are working in tandem to reduce the "Greenhouse Effect" and transition to a sustainable economy.

1. Understanding the Mechanism (The Science)

Science provides the diagnostic tools to understand the problem. By analyzing ice cores and satellite data, scientists have identified that the concentration of CO₂ has increased significantly since the industrial era.

The "Greenhouse Effect" occurs when gases like carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O) trap heat in the atmosphere that would otherwise escape into space.

2. Technological Solutions for Mitigation

Technology focuses on "Mitigation"—the active reduction of greenhouse gas emissions.

Renewable Energy Systems: Engineers are replacing coal and gas with Solar Photovoltaics (PV) and Wind Turbines. Modern technology has made these sources cheaper than fossil fuels in most parts of the world.

Green Hydrogen: Electrolyzers use renewable electricity to split water (H₂O) into hydrogen and oxygen. This "Green Hydrogen" can power heavy industries like steel and cement that cannot easily run on batteries.

Carbon Capture and Storage (CCS): This involves "scrubbing" CO₂ directly from factory chimneys or the open air (Direct Air Capture). The captured gas is then compressed and injected into deep underground rock formations for permanent storage.

Q-2. Discuss the impact of climate change on ecology.

Ans - Climate change is one of the most significant challenges facing global ecosystems today, affecting everything from individual species' behavior to the stability of entire biomes. Addressing these shifts requires a combination of aggressive mitigation and a transition toward sustainable development.

Impact of Climate Change on Ecology

The ecological impact of climate change is multifaceted, often triggering "tipping points" that can lead to irreversible changes in the natural world.

Loss of Biodiversity and Habitat: As temperatures rise, many species are forced to migrate to higher latitudes or altitudes to find suitable climates. Species that cannot move or adapt, such as those in polar regions or on mountain peaks, face extinction.

Phenological Mismatches: Climate change alters the timing of biological events. For example, plants may bloom earlier in the spring, but the insects that pollinate them may not emerge at the same time, disrupting critical food chains and reproductive cycles.

Ocean Acidification and Warming: Increased CO₂ absorption makes seawater more acidic, which hinders the ability of marine organisms like corals and shellfish to form shells (calcification). Rising temperatures also lead to coral bleaching, destroying some of the most biodiverse habitats on Earth.

Shift in Ecosystem Boundaries: We are seeing the "browning" of tropical forests and the "greening" of Arctic tundras. These shifts alter the carbon sequestration capacity of the land, potentially turning carbon sinks into carbon sources.

THANK
you!

The image features the words "THANK you!" written in a warm, brown, hand-drawn script font. The word "THANK" is in all caps with a tall, elegant 'T' that curves over the 'H'. "you!" is in lowercase, with a long, sweeping tail on the 'y' that curves under the 'u'. The text is set against a light brown, textured background that resembles aged paper. Surrounding the text are various decorative elements: several green leaves with visible veins, small orange flowers with five petals, and tiny orange four-pointed stars. Some small brown dots are also scattered around the text, adding to the rustic, hand-drawn feel of the illustration.